BUSINESS INTELLIGENCE CASE STUDY

NexaTel Communications
Customer Churn & Retention Analysis

Business Brief & Key Performance Indicator Framework

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Analysis Period: January 2023 – December 2025

Dataset Scope: ~15,000 customers · 4 linked tables · ~450,000 records

Tools: MySQL (data cleaning & analysis) · Power BI (dashboard & modeling)

Portfolio Project — NexaTel Communications is a simulated telecom dataset built for demonstrating end-to-end analytics capability, including SQL data cleaning, dimensional modeling, and business intelligence dashboard design.

1. Business Context

NexaTel Communications is a telecom service provider serving approximately 15,000 customers. Over the three-year period from January 2023 to December 2025, the business has experienced a meaningful volume of customer churn, and leadership has requested an analysis to understand the underlying drivers, quantify the financial exposure, and evaluate whether current retention efforts are working.

> **The Business Question:** Why are customers leaving NexaTel, what is churn costing the business in revenue, and are the company's retention campaigns delivering a positive return on investment?

This brief translates that question into a concrete analytical scope, a data model, and a set of KPIs that the dashboard is designed to answer. It is intended to give any stakeholder — technical or non-technical — a clear view of what the analysis covers and why each metric matters before they look at the dashboard itself.

2. Objectives

- **Diagnose churn:** Identify which customer segments, contract types, and service conditions are most associated with churn.
- **Quantify impact:** Translate churn into revenue terms — how much recurring revenue is at risk, and what is the lifetime value being lost.
- **Evaluate retention:** Determine whether retention campaigns are reaching the right customers and whether the cost of running them is justified by the revenue they save.
- **Recommend action:** Surface clear, data-backed recommendations that the business could act on to reduce churn going forward.

3. Data Sources & Model

Four linked tables underpin the analysis, covering the full customer lifecycle:

TABLE	GRAIN	ROWS	CAPTURES
Customer_Master	1 row per customer	~15,000	Customer profile, contract/service details, billing setup, and final churn outcome
Monthly_Customer_Activity	1 row per customer per active month	~300,000	Usage, network quality, billing, payment behavior, and competitor pressure signals, tracked month over month
Customer_Support_Interactions	1 row per support interaction	~59,000	Support channel, issue type, and resolution outcome
Retention_Campaigns	1 row per campaign event	~28,000	Retention offers extended to customers, their response, and the cost of the offer

Churn Lifecycle Logic

```
Customer_Master  →  holds the final outcome (Churn_Status, Churn_Date, Churn_Reason, Churn_Category)
Monthly_Customer_Activity  →  tracks status month-by-month; churn month = final record, status flips to
"Churned," no activity generated after
Active customers  →  NULL churn fields, continuous monthly records through their last active month
```

Churn likelihood is driven by a combination of tenure, contract type, network quality, complaint history, satisfaction, payment issues, and competitor pressure — with natural randomness layered in so outcomes are realistic rather than deterministic.

Power BI Data Model

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Dim_Customer  &  Dim_Date  →  Fact_Monthly_Activity
                               →  Fact_Support_Interactions
                               →  Fact_Retention_Campaigns
```

A standard star schema, with DAX measures layered on top for time intelligence and the KPIs below.

4. Data Quality Notes

The raw source data intentionally contains realistic data quality issues, consistent with what an analyst would encounter with real operational exports: duplicate records, missing values, inconsistent casing and category labels (e.g. contract type, Yes/No fields), mixed date formats, incorrect data types, and a small number of outliers. All cleaning is performed in MySQL prior to modeling, with raw and cleaned tables kept separate to preserve an auditable trail.

5. Dashboard Structure

The dashboard is organized into four pages, each answering a different layer of the business question:

#	PAGE	ANSWERS
1	Executive Churn Overview	What is happening — headline churn and business health metrics
2	Customer & Churn Drivers	Why customers are leaving — segment and behavioral drivers
3	Revenue & Customer Value	What it's costing the business — financial impact of churn
4	Retention & Recommendations	What to do about it — campaign effectiveness and next steps

6. Key Performance Indicators

KPIs are grouped into four categories, aligned to the dashboard pages above.

GROUP	KPIS
CUSTOMER	Churn rate · Retention rate · Average customer tenure
REVENUE	Monthly Recurring Revenue (MRR) · Average Revenue Per User (ARPU) · Revenue at risk · Customer Lifetime Value (CLV)
SERVICE	Network quality score · Customer satisfaction · Complaint rate
RETENTION	Campaign success rate · Retention cost vs. revenue saved

7. Intended Outcome

By the end of this analysis, the dashboard should let a stakeholder answer, at a glance: who is churning and why, how much revenue that represents, and whether NexaTel's retention spend is producing a positive return. The final page translates these findings into concrete, prioritized recommendations for reducing churn and improving retention ROI going forward.

This document is a business brief for a self-directed portfolio project. The underlying dataset was synthetically generated to reflect realistic telecom churn dynamics and data quality conditions for the purpose of demonstrating SQL data cleaning, dimensional modeling, and BI dashboard design skills.